Fiscal Year:	FY 2015	**Task Last Updated:**	FY 01/25/2022
PI Name:	Alptekin, Gokhan Ph.D.		
Project Title:	A Low-Power Medical Oxygen Generator		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	galptekin@tda.com	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	303-940-2349
Organization Name:	TDA Research, Inc.		
PI Address 1:	12345 West 52nd Avenue		
PI Address 2:			
PI Web Page:			
City:	Wheat Ridge	**State:**	CO
Zip Code:	80033-1916	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	SBIR Phase II
Start Date:	05/21/2015	**End Date:**	05/20/2017
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX15CC17C		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	An on-board oxygen concentrator is required during long duration crewed space missions to supply medical oxygen. The commercial medical oxygen generators based on pressure swing adsorption (PSA) are large and highly power intensive. TDA Research, Inc. (TDA) proposes to develop a small, lightweight, portable oxygen generator based on a vacuum swing adsorption (VSA) to produce concentrated medical oxygen. The unit uses ambient vehicle cabin air as the feed and delivers high purity oxygen while meeting NASA's requirements for high flow capacity, closed-loop tissue oxygen control and operation in microgravity/partial gravity. TDA's VSA system uses a modified version of the lithium exchanged low silica X zeolite (Li-LSX), a state-of-the-art air separation sorbent extensively used in commercial Portable Oxygen Concentrators (POCs) to enhance the N2 adsorption capacity. In Phase I, we demonstrated the scientific, technical, and commercial feasibility of the oxygen concentrator module (OCM). In Phase II, we will develop a higher fidelity prototype with an adjustable pressure output to produce 2-15 lpm of O2 at 50% to +90% purity from ambient cabin air. The OCM will be capable of self-regulating the oxygenation of the patient using a closed loop		

	feedback system that senses tissue oxygenation level. We will evaluate the sorbent performance in a breadboard bench-scale prototype under simulated microgravity/partial gravity exploration atmospheres and carry out a 1,500 hr longevity test (at a minimum) to determine its mechanical durability. Based on the experimental results, we will design a prototype unit that will meet all of NASA's requirements (e.g., low power draw over the range of flows and oxygen levels, lightweight and volume), while delivering the desired oxygen flow and purity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New project for FY2015. Reporting not required for this SBIR Phase II project. (Ed. note: added to Task Book on 01/20/2022)
Bibliography Type:	Description: (Last Updated:)